

PROFIT & LOSS

Complete Theory, Solved Examples & Comprehensive Practice Set (English)

1. THEORY & CONCEPTS

The topic **Profit and Loss** deals with commercial mathematics calculations regarding buying and selling items in business markets.

Key Terminology:

- **Cost Price (CP):** The price at which an article is purchased.
- **Selling Price (SP):** The price at which an article is sold.
- **Profit (Gain):** When $SP > CP$, then $Profit = SP - CP$
- **Loss:** When $CP > SP$, then $Loss = CP - SP$

Important Formulas:

1. **Profit %:** $Profit \% = (Profit / CP) \times 100$
2. **Loss %:** $Loss \% = (Loss / CP) \times 100$
3. **Selling Price from CP and Profit %:** $SP = CP \times ((100 + Profit \%) / 100)$
4. **Selling Price from CP and Loss %:** $SP = CP \times ((100 - Loss \%) / 100)$
5. **Cost Price from SP and Profit %:** $CP = (SP \times 100) / (100 + Profit \%)$
6. **Cost Price from SP and Loss %:** $CP = (SP \times 100) / (100 - Loss \%)$

2. SOLVED EXAMPLES (12 Examples)

Q1. A shopkeeper buys an article for ₹400 and sells it for ₹480. Find the profit percentage.

Solution: $CP = ₹400$, $SP = ₹480$

$Profit = 480 - 400 = ₹80$

$Profit \% = (80 / 400) \times 100 = 20\%$

Answer: 20%

Q2. An item is bought for ₹750 and sold at a loss of 12%. Find the selling price.

Solution: CP = ₹750, Loss % = 12%

$$SP = CP \times ((100 - 12) / 100) = 750 \times (88 / 100) = ₹660$$

Answer: ₹660

Q3. If the selling price of an article is ₹920 and the profit is 15%, find its cost price.

Solution: SP = ₹920, Profit % = 15%

$$CP = (SP \times 100) / (100 + \text{Profit \%}) = (920 \times 100) / 115 = ₹800$$

Answer: ₹800

Q4. A man sold a cycle for ₹1200 at a loss of 20%. At what price should he sell it to gain 10%?

Solution: CP = $(1200 \times 100) / (100 - 20) = (1200 \times 100) / 80 = ₹1500$

$$\text{New SP for 10\% gain} = 1500 \times (110 / 100) = ₹1650$$

Answer: ₹1650

Q5. If the cost price of 15 articles is equal to the selling price of 12 articles, find the profit percentage.

Solution: Let CP of 1 article be ₹1. CP of 15 articles = ₹15. SP of 12 articles = ₹15.

$$\text{Profit} = 15 - 12 = 3 \text{ on 12 articles. Profit \%} = (3 / 12) \times 100 = 25\%$$

Answer: 25%

Q6. A trader mixes 26 kg of rice at ₹20 per kg with 30 kg of rice at ₹36 per kg. At what price per kg should he sell the mixture to gain 10%?

Solution: Total cost = $(26 \times 20) + (30 \times 36) = 520 + 1080 = ₹1600$ for 56 kg.

$$\text{Average CP per kg} = 1600 / 56 = ₹28.57. \text{ SP for 10\% gain} = 28.57 \times 1.10 = ₹31.43$$

Answer: ₹31.43 per kg

Q7. A shopkeeper sells an article at a profit of 10%. Had he bought it for 4% less and sold it for ₹18 more, he would have gained $18\frac{3}{4}\%$. Find the cost price.

Solution: Let initial CP = $100x$, SP = $110x$. New CP = $96x$. New SP = $96x \times (75 / 4) / 100 + 96x = 114x$.

Difference = $114x - 110x = 4x = 18 \Rightarrow x = 4.5$. CP = $100 \times 4.5 = ₹450$

Answer: ₹450

Q8. By selling 33 meters of cloth, a person gains the selling price of 11 meters. Find the gain percentage.

Solution: Gain = SP of 33m - CP of 33m = SP of 11m $\Rightarrow$ CP of 33m = SP of 22m.

Profit % = $((33 - 22) / 22) \times 100 = (11 / 22) \times 100 = 50\%$

Answer: 50%

Q9. A dishonest dealer professes to sell his goods at cost price, but uses a weight of 900 grams instead of 1 kg. Find his actual profit percentage.

Solution: Profit % = $((\text{Error} / (\text{True Value} - \text{Error})) \times 100) = ((100 / 900) \times 100) = 11.11\%$

Answer: 11.11%

Q10. A person sells two articles for ₹990 each. On one he gains 10% and on the other he loses 10%. Find the overall profit or loss percentage.

Solution: When SP is same and profit/loss % is equal (x), there is always a loss of $(x^2 / 100)\%$.

Loss % = $(10^2 / 100)\% = 1\% \text{ loss}$

Answer: 1% loss

Q11. If goods are valued at ₹6000 and a discount of 20% is offered, followed by a further cash discount of 5%, what is the final selling price?

Solution: First SP = $6000 \times 0.80 = ₹4800$. Final SP = $4800 \times 0.95 = ₹4560$

Answer: ₹4560

Q12. A clock is sold at a 5% profit. If it is sold for ₹60 more, the profit is 8%. Find the cost price of the clock.

Solution: Difference in percentage = $8\% - 5\% = 3\%$

$3\% = ₹60 \Rightarrow$ CP (100%) = $(60 / 3) \times 100 = ₹2000$

Answer: ₹2000

3. PRACTICE QUESTIONS (25 Questions)

Q1. An article bought for ₹500 is sold for ₹600. Find the profit percentage.

- (a) 15% (b) 20% (c) 25% (d) 10%

Q2. A radio is purchased for ₹800 and sold at a loss of 10%. Find the selling price.

- (a) ₹700 (b) ₹720 (c) ₹750 (d) ₹780

Q3. If the selling price of an article is ₹420 and the profit is 5%, what is the cost price?

- (a) ₹390 (b) ₹400 (c) ₹410 (d) ₹380

Q4. The cost price of 12 pens is equal to the selling price of 8 pens. Find the profit percentage.

- (a) 33.33% (b) 50% (c) 40% (d) 25%

Q5. A man sells an article at a profit of 20%. If he had bought it at 20% less and sold it for ₹75 less, he would have gained 25%. Find the cost price.

- (a) ₹300 (b) ₹350 (c) ₹400 (d) ₹450

Q6. If a person incurs a loss of 5% by selling an article for ₹1140, at what price should he sell it to make a 5% profit?

- (a) ₹1200 (b) ₹1260 (c) ₹1300 (d) ₹1350

Q7. By selling 25 meters of cloth, a trader gains the selling price of 5 meters. Find the profit percentage.

- (a) 20% (b) 25% (c) 30% (d) 15%

Q8. A dishonest shopkeeper promises to sell his goods at cost price but uses a 920g weight instead of 1 kg. Find his gain percentage.

- (a) 8% (b) 8.7% (c) 9.1% (d) 10%

Q9. Two horses are sold for ₹1920 each. On one, the seller gains 20% and on the other, he loses 20%. What is the net outcome?

- (a) No Profit, No Loss (b) 4% Loss (c) 4% Profit (d) 2% Loss

Q10. A shopkeeper marks his goods 20% above the cost price and allows a discount of 10% on marked price. Find his profit percentage.

- (a) 8% (b) 10% (c) 12% (d) 15%

Q11. If the cost price of 50 articles equals the selling price of 40 articles, find the profit percent.

- (a) 20% (b) 25% (c) 30% (d) 40%

Q12. A book is sold for ₹270 at a profit of 12.5%. Find the cost price of the book.

- (a) ₹220 (b) ₹230 (c) ₹240 (d) ₹250

Q13. A person sold an article at a loss of 10%. If he had sold it for ₹90 more, he would have gained 5%. Find the cost price of the article.

- (a) ₹500 (b) ₹600 (c) ₹700 (d) ₹800

Q14. If a discount of 15% is allowed on the marked price of an article, it is sold for ₹425. Find the marked price.

- (a) ₹480 (b) ₹500 (c) ₹520 (d) ₹550

Q15. A dealer allows a 10% discount on his marked price and still gains 20%. If the marked price is ₹800, find the cost price.

- (a) ₹550 (b) ₹600 (c) ₹650 (d) ₹700

Q16. The difference between a discount of 30% and two successive discounts of 20% and 10% on a bill of ₹2000 is:

- (a) ₹0 (b) ₹20 (c) ₹40 (d) ₹50

Q17. A vendor bought bananas at 6 for ₹5 and sold them at 5 for ₹6. Find his profit percentage.

- (a) 30% (b) 35% (c) 44% (d) 50%

Q18. By selling an article for ₹144, a man loses as much percent as the cost price of the article. Find the cost price.

- (a) ₹160 or ₹90 (b) ₹180 or ₹80 (c) ₹200 or ₹70 (d) ₹150 or ₹90

Q19. A man sold two chairs for ₹1200 each. One was sold at 25% profit and the other at 25% loss. Find the overall loss in rupees.

- (a) ₹100 (b) ₹150 (c) ₹160 (d) ₹200

Q20. If the manufacturer gains 10%, the wholesale dealer 15%, and the retailer 25%, then find the cost of production of an article if its retail price is ₹1265.

(a) ₹750 (b) ₹800 (c) ₹850 (d) ₹900

Q21. A trader lists his goods by 30% above the cost price and allows a 15% discount. What is his profit percentage?

(a) 10.5% (b) 12% (c) 15% (d) 18.5%

Q22. If an article is sold at 20% profit instead of 10% loss, the seller gets ₹70 more. Find the cost price.

(a) ₹300 (b) ₹350 (c) ₹400 (d) ₹450

Q23. A person sells a table at a profit of 10%. If he had bought it at 5% less and sold it for ₹80 more, he would have gained 20%. Find the CP.

(a) ₹1500 (b) ₹1600 (c) ₹1800 (d) ₹2000

Q24. A milkman mixes 20 litres of water with 80 litres of pure milk. If he sells the mixture at the cost price of pure milk, find his profit percentage.

(a) 20% (b) 25% (c) 30% (d) 33.33%

Q25. If the successive discounts of 20%, 10%, and 5% are given, find the single equivalent discount percentage.

(a) 31.6% (b) 35% (c) 30.4% (d) 32.4%

4. ANSWER KEY

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
1	(b) 20%	10	(a) 8%	19	(c) ₹160
2	(b) ₹720	11	(b) 25%	20	(a) ₹750
3	(b) ₹400	12	(c) ₹240	21	(a) 10.5%
4	(b) 50%	13	(b) ₹600	22	(b) ₹350
5	(a) ₹300	14	(b) ₹500	23	(b) ₹1600
6	(b) ₹1260	15	(b) ₹600	24	(b) 25%
7	(a) 20%	16	(b) ₹20	25	(a) 31.6%
8	(b) 8.7%	17	(c) 44%		
9	(b) 4% Loss	18	(b) ₹180 or ₹80		